1.03 Coordinate Geometry: Linear Graphs

Worked Solutions

1 a)

$m = \frac{7-5}{1-2} = -2$ so $y - 5 = -2(x - 2)$

$\therefore 2x + y = 9$

1 b)

$m = \frac{13-4}{4-1} = 3$ so $y - 4 = 3(x - 1)$

$\therefore -3x + y = 1$

1 c)

$m = \frac{1+3}{5+1} = \frac{2}{3}$ so $y + 3 = \frac{2}{3}(x + 1)$

$\therefore -2x + 3y = -7$

1 d)

$m = \frac{-17+6}{-2+4} = -\frac{11}{2}$ so $y + 6 = -\frac{11}{2}(x + 4)$

$\therefore 11x + 2y = -56$

1 e)

$m = \frac{4q-q}{2p-p} = \frac{3q}{p}$ so $y - q = \frac{3q}{p}(x - p)$

$\Rightarrow py - pq = 3qx - 3pq$

$\therefore 3qx - py = 2pq$

1 f)

$m = \frac{p-\frac{1}{p}}{\frac{1}{p}-p} = \frac{p^2-1}{1-p^2} = -\frac{p^2-1}{p^2-1}$ so $y - \frac{1}{p} = -(x - p)$

$\Rightarrow y - \frac{1}{p} = -x + p \Rightarrow y + x = p + \frac{1}{p}$

$\therefore px + py = p^2 + 1$

2 a)

$6y = -3x + 7 \Rightarrow y = -\frac{1}{2}x + \frac{7}{2}$

$m_1 = -\frac{1}{2}$ and $m_2 = 2$

2 b)

$3y = 2x + 1 \Rightarrow y = \frac{2}{3}x + \frac{1}{3}$

$m_1 = \frac{2}{3}$ and $m_2 = -\frac{3}{2}$

2 c)

$8y = -2x + 11 \Rightarrow y = -\frac{1}{4}x + \frac{11}{8}$

$m_1 = -\frac{1}{4}$ and $m_2 = 4$

2 d)

$7y = 5x + 2 \Rightarrow y = \frac{5}{7}x + \frac{2}{7}$

$m_1 = \frac{5}{7}$ and $m_2 = -\frac{7}{5}$

2 e)

$17y = -20x + 3 \Rightarrow y = -\frac{20}{17}x + \frac{3}{17}$

$m_1 = -\frac{20}{17}$ and $m_2 = \frac{17}{20}$

2 f)

$by = -ax + c \Rightarrow y = -\frac{a}{b}x + \frac{c}{b}$

$m_1 = -\frac{a}{b}$ and $m_2 = \frac{b}{a}$

3 a)

$2x + 3y = 1 \Rightarrow y = -\frac{2}{3}x + \frac{1}{2}$

$-\frac{2}{3} \times -\frac{3}{2} \neq -1 \therefore$ not perpendicular.

3 b)

$x - 5y - 10 = 0 \Rightarrow y = \frac{1}{5}x + 2$

$\frac{1}{5} \times 5 \neq -1 \therefore$ not perpendicular.

3 c)

$24x + 12y = 16 \Rightarrow y = -2x + \frac{4}{3}$

$12y + 24x = 1 \Rightarrow y = -2x + \frac{1}{12}$

$-2 \times -2 \neq -1 \therefore$ not perpendicular (they are parallel).

3 d)

$\frac{x}{2} + \frac{y}{8} = 1 \Rightarrow y = -4x + 8$

$4y - x = 28 \Rightarrow y = \frac{1}{4}x + 7$

$-4 \times \frac{1}{4} = -1 \therefore$ perpendicular.

3 e)

$ax + by = p \Rightarrow y = -\frac{a}{b}x + \frac{p}{b}$

$bx - ay = q \Rightarrow y = \frac{b}{a}x - \frac{q}{a}$

$-\frac{a}{b} \times \frac{b}{a} = -1 \therefore$ perpendicular.

4 a)

$ax + by = c \Rightarrow y = -\frac{a}{b}x + \frac{c}{b}$

$px + qy = r \Rightarrow y = -\frac{p}{q}x + \frac{r}{q}$

$\frac{a}{b} = \frac{p}{q}$ or $aq = bp$

4 b)

$ax + by = c \Rightarrow y = -\frac{a}{b}x + \frac{c}{b}$

$px + qy = r \Rightarrow y = -\frac{p}{q}x + \frac{r}{q}$

$-\frac{a}{b} \times -\frac{p}{q} = -1$

$\frac{ap}{bq} = -1$ or $ap = -bq$

5 a)

$(\frac{-4+4}{2}, \frac{3+9}{2}) = (0, 6)$

5 b)

$(\frac{1+6}{2}, \frac{7-5}{2}) = (\frac{7}{2}, 1)$

5 c)

$(\frac{-8-2}{2}, \frac{10-1}{2}) = (-5, \frac{9}{2})$

5 d)

$(\frac{-7+\frac{7}{2}}{2}, \frac{-2+12}{2}) = (-\frac{7}{4}, 5)$

5 e)

$(\frac{p+2p}{2}, \frac{q+3q}{2}) = (\frac{3}{2}p, 2q)$

5 f)

$(\frac{p+\frac{1}{p}}{2}, \frac{p+\frac{1}{p}}{2}) = (\frac{1}{2}p + \frac{1}{2p}, \frac{1}{2}p + \frac{1}{2p})$

6 a)

$2 \times 4 - 5k = 16 \Rightarrow k = -\frac{8}{5}$

6 b)

$4(2k) - 3(k) + 20 = 0 \Rightarrow 5k = -20$

$\Rightarrow k = -4$

7 a)

$m_1 = \frac{1-5}{6+2} = -\frac{1}{2}$ midpoint: $(\frac{-2+6}{2}, \frac{5+1}{2}) = (2, 3)$

$y - 3 = 2(x - 2) \Rightarrow y - 3 = 2x - 4$

$\Rightarrow 2x - y = 1$

7 b)

$m_1 = \frac{8-5}{6-3} = 1$ midpoint: $(\frac{3+6}{2}, \frac{5+8}{2}) = (\frac{9}{2}, \frac{13}{2})$

$y - \frac{13}{2} = -1(x - \frac{9}{2}) \Rightarrow y - \frac{13}{2} = -x + \frac{9}{2}$

$\Rightarrow x + y = 11$

7 c)

$m_1 = \frac{10-2}{4-1} = \frac{8}{3}$ midpoint: $(\frac{1+4}{2}, \frac{2+10}{2}) = (\frac{5}{2}, 6)$

$y - 6 = -\frac{3}{8}(x - \frac{5}{2}) \Rightarrow -8y + 48 = 3x - \frac{15}{2}$

$\Rightarrow 8y + 3x = \frac{111}{2} \Rightarrow 6x + 16y = 111$

7 d)

$m_1 = \frac{-5-2}{-1+6} = -\frac{7}{5}$ midpoint: $(\frac{-6-1}{2}, \frac{2-5}{2}) = (-\frac{7}{2}, -\frac{3}{2})$

$y + \frac{3}{2} = \frac{5}{7}(x + \frac{7}{2}) \Rightarrow 7y + \frac{21}{2} = 5x + \frac{35}{2}$

$\Rightarrow 5x - 7y = -7$

7 e)

$m_1 = \frac{-2-11}{-1+9} = -\frac{13}{8}$ midpoint: $(\frac{-9-1}{2}, \frac{11-2}{2}) = (-5, \frac{9}{2})$

$y - \frac{9}{2} = \frac{8}{13}(x + 5) \Rightarrow 13y - \frac{117}{2} = 8x + 40$

$\Rightarrow 13y - 8x = \frac{197}{2} \Rightarrow 16x - 26y = -197$

7 f)

$m_1 = \frac{3q-q}{2p-p} = \frac{2q}{p}$ midpoint: $(\frac{p+2p}{2}, \frac{q+3q}{2}) = (\frac{3}{2}p, 2q)$

$y - 2q = -\frac{p}{2q}(x - \frac{3}{2}p) \Rightarrow 2qy - 4q^2 = -px + \frac{3}{2}p^2$

$\Rightarrow 4qy - 8q^2 = -2px + 3p^2 \Rightarrow 2px + 4qy = 8q^2 + 3p^2$

8)

$y = \frac{1}{2}x + 5$ and $(-1, 4)$

$y - 4 = \frac{1}{2}(x + 1) \Rightarrow 2y - 8 = x + 1$

$\Rightarrow x - 2y = -9$

9)

$y = -\frac{1}{3}x - \frac{11}{6}$ and $(2, 3)$

$y - 3 = 3(x - 2) \Rightarrow y - 3 = 3x - 6$

$\Rightarrow 3x - y = 3$

10 a)

Angle between $y = 0$ and the x-axis: $0°$

Angle between $y = 2x$ and the x-axis: $\tan^{-1}(\frac{2}{1}) = 63.4°$

Angle between lines: $63.4° - 0° = 63.4°$

10 b)

Angle between $y = 3x + 3$ and the x-axis: $\tan^{-1}(\frac{3}{1}) = 71.6°$

Angle between $y = 4x - 1$ and the x-axis: $\tan^{-1}(\frac{4}{1}) = 76.0°$

Angle between lines: $76.0° - 71.6° = 4.4°$

10 c)

Angle between $y = 5x + 8$ and the x-axis: $\tan^{-1}(\frac{5}{1}) = 78.7°$

Angle between $y = -x$ and the x-axis: $90° + \tan^{-1}(\frac{1}{1}) = 135°$

Angle between lines: $135° - 78.7° = 56.3°$

10 d)

Angle between $y = -2x - 4$ and the x-axis: $90° + \tan^{-1}(\frac{2}{1}) = 153.4°$

Angle between $y = -3x + 1$ and the x-axis: $90° + \tan^{-1}(\frac{5}{1}) = 168.7°$

Angle between lines: $168.7° - 153.4° = 15.3°$

11)

Meets the axes at $(0, \frac{11}{2})$ and $(11, 0)$

Area of triangle $= \frac{1}{2} \times \frac{11}{2} \times 11 = \frac{121}{4}$ units^2

12)

Meets the axes at $(0, \frac{2}{k})$ and $(\frac{2}{3}, 0)$

Area of triangle $= \frac{1}{2} \times \frac{2}{k} \times \frac{2}{3} = \frac{2}{3k}$ units^2

13)

$m_1 = \frac{-2-11}{4-3} = -13$

$m_2 = \frac{1+2}{-2-4} = -\frac{3}{6} = -\frac{1}{2}$

$m_3 = \frac{1-11}{-2-3} = \frac{10}{5} = 2$

So the line passing through $(4, -2)$ and $(-2, 1)$ is perpendicular to the line passing through $(3, 11)$ and $(2, -1)$ so it must be a right-angled triangle.

14)

$$\sqrt{(-1-5)^2 + (8-10)^2} = \sqrt{36+4} = \sqrt{40}$$

$$\sqrt{(4+1)^2 + (-7-8)^2} = \sqrt{25+225} = \sqrt{250}$$

$$\sqrt{(4-5)^2 + (-7-10)^2} = \sqrt{1+289} = \sqrt{290}$$

$(\sqrt{40})^2 + (\sqrt{250})^2 = (\sqrt{290})^2 \therefore$ a right-angled triangle.

Area: $\frac{1}{2} \times \sqrt{40} \times \sqrt{250} = 50$ units^2

15)

$y = \frac{7}{2}x$ and $y = -\frac{6}{k}x + \frac{10}{k}$

So $\frac{7}{2} = \frac{k}{6} \Rightarrow k = 21$

16)

$y = -2x + 9$ so gradient is $\frac{1}{2}$

$\frac{7-6}{1-p} = \frac{1}{2} \Rightarrow 1 - p = 2 \Rightarrow p = -1$

17)

Midpoint: $(\frac{p+5}{2}, \frac{q-2}{2})$

Midpoint in perpendicular bisector:

$3(\frac{q-2}{2}) + (\frac{p+5}{2}) = 15 \Rightarrow 3(q-2) + p + 5 = 30$

$\Rightarrow p + 3q = 31$

Gradients:

$y = -\frac{1}{3}x + 5$ so $3 = \frac{q+2}{5-p} \Rightarrow 3(5-p) = q + 2$

$\Rightarrow 15 - 3p = q + 2 \Rightarrow 3p + q = 13$

Solving: $p = 1, q = 10$

18)

$m_1 = \frac{13-10}{10+5} = \frac{3}{15} = \frac{1}{5}$ and $m_2 = \frac{5-10}{-15-10} = \frac{5}{25} = \frac{1}{5}$

Since both gradients are the same and they have a common point, then they are collinear.

19 a)

$m_1 = \frac{3-9}{3+15} = \frac{6}{-18} = -\frac{1}{3}$ and $m_2 = \frac{-5-3}{k-3} = -\frac{8}{k-3}$

Since both gradients are the same

$-\frac{1}{3} = -\frac{8}{k-3} \Rightarrow k - 3 = 24 \Rightarrow k = 27$

19 b)

$m_1 = \frac{3-k}{-3k-1}$ and $m_2 = \frac{\frac{3}{2}-3}{7+3k}$

Since both gradients are the same

$\frac{3-k}{-3k-1} = \frac{\frac{3}{2}-3}{7+3k} \Rightarrow (3-k)(7+3k) = -\frac{3}{2}(-3k-1)$

$\Rightarrow -3k^2 + 2k + 21 = \frac{9}{2}k + \frac{3}{2}$

$\Rightarrow -3k^2 - \frac{5}{2}k + \frac{39}{2} = 0$ so $\Rightarrow k = -3$ or $k = \frac{13}{6}$

20 a)

$m_1 = \frac{2-10}{5+3} = -1$ $m_2 = 1$

$(\frac{5-3}{2}, \frac{2+10}{2}) = (1, 6)$

$y - 6 = 1(x-1) \Rightarrow y = x + 5$

20 b)

$m_1 = \frac{2-6}{5-9} = 1$ $m_2 = -1$

$(\frac{5+9}{2}, \frac{2+6}{2}) = (7, 4)$

$y - 4 = -1(x-7) \Rightarrow y = -x + 11$

20 c)

$x + 5 = -x + 11 \Rightarrow x = 3, y = 8$

20 d)

AP: $\sqrt{(3+3)^2+(8-10)^2} = \sqrt{36+4} = 2\sqrt{10}$

BP: $\sqrt{(3-5)^2+(8-2)^2} = \sqrt{4+36} = 2\sqrt{10}$

CP: $\sqrt{(3-9)^2+(8-6)^2} = \sqrt{36+4} = 2\sqrt{10}$

21)

$y = \frac{4}{3}x + \frac{5}{3}$ so $m = -\frac{3}{4}$

$y - 8 = -\frac{3}{4}(x-11) \Rightarrow y = -\frac{3}{4}x + \frac{65}{4}$

Point of intersection: $\frac{4}{3}x + \frac{5}{3} = -\frac{3}{4}x + \frac{65}{4}$

$\Rightarrow \frac{175}{12} = \frac{25}{12}x \Rightarrow x = 7, y = 11$

Distance between points: $\sqrt{(11-7)^2+(8-11)^2} = \sqrt{16+9} = 5$

22 a)

$m_{AB} = \frac{4-10}{5-10} = \frac{6}{5}$ and $m_{BC} = \frac{1-4}{20-5} = -\frac{1}{5}$

$m_{AB}m_{BC} \neq -1$ so not perpendicular

22 b)

$y - 4 = -\frac{1}{5}(x-5) \Rightarrow y = -\frac{1}{5}x + 5$

22 c)

Gradient is $-\frac{1}{5}$ so $y - 10 = -\frac{1}{5}(x-10)$

$\Rightarrow y = -\frac{1}{5}x + 12$

22 d)

$k = -\frac{1}{5} \times 15 + 12 = 9$

22 e)

AD: $\sqrt{(15-10)^2+(9-10)^2} = \sqrt{25+1} = \sqrt{26}$

BC: $\sqrt{(1-4)^2+(20-5)^2} = \sqrt{9+225} = 3\sqrt{26}$

22 f)

$m = 5$ so $y - 10 = 5(x-10) \Rightarrow y = 5x - 40$

22 g)

Intersection:

$-\frac{1}{5}x + 5 = 5x - 40 \Rightarrow \frac{26}{5}x = 45$

$\Rightarrow x = \frac{225}{26}, y = \frac{85}{26}$

$\sqrt{(10-\frac{225}{26})^2+(10-\frac{85}{26})^2} = \sqrt{\frac{1225}{676}+\frac{30625}{676}} = \frac{35}{26}\sqrt{26}$

22 h)

$\frac{1}{2}(\frac{35}{26}\sqrt{26})(3\sqrt{26}+\sqrt{26}) = 70$ units^2

23 a)

Distance AB: $\sqrt{(3-2)^2+(1-(-6))^2} = 5\sqrt{2}$

Distance BC: $\sqrt{(3-8)^2+(1-(-4))^2} = 5\sqrt{2}$

$\therefore$ distances are the same

23 b)

Line going through AC:

$y-2 = \frac{8-2}{-4-(-6)}(x-6) \Rightarrow y-2 = 3(x+6)$

Line going through BD:

$y-3 = -\frac{1}{3}(x-1) \Rightarrow 3y-9 = -x+1$

$\Rightarrow 3y+x = 10$ as req.

24)

Midpoint of AC: $(\frac{-3+4}{2}, \frac{-1+4}{2}) = (\frac{1}{2}, \frac{3}{2})$

Midpoint of BD: $(\frac{2-1}{2}, \frac{-\frac{3}{5}+\frac{18}{5}}{2}) = (\frac{1}{2}, \frac{3}{2})$

Gradient of AC: $m = \frac{4-(-1)}{4-(-3)} = \frac{5}{7}$

Gradient of BD: $m = \frac{\frac{18}{5}-(-\frac{3}{5})}{-1-2} = -\frac{7}{5}$

$\therefore AC$ and BD are perpendicular bisectors of each other

Distance AC: $\sqrt{(-1-4)^2+(-3-4)^2} = \sqrt{74}$

Distance BD: $\sqrt{(\frac{18}{5}+\frac{3}{5})^2+(-1-2)^2} = \frac{3}{5}\sqrt{74}$

As the distances are different, it is not a square.

So it must be a rhombus.